

5. Project Development Phase

ComicCraft – AI Comic Story Creator

Introduction

The development phase implements the design as a complete Python-based web application in Visual Studio Code.

Backend Development

FastAPI and Uvicorn form the backend foundation. The backend provides the website route, health check, story-generation endpoint, complete comic-generation endpoint, panel-regeneration endpoint, and comic retrieval endpoint.

Gemini Integration

The Gemini service is responsible for generating original stories with consistent characters and logical panel continuity. Its response is expected to contain title, summary, character profile, panels, narration, dialogue, scene descriptions, and image prompts. Pydantic validation is used to check the generated structure.

Image Generation

The image service generates panel illustrations using Hugging Face Diffusers/Stable Diffusion and stores them in generated/images/. Prompts include character appearance, action, environment, lighting, camera angle, mood, and art style.

PDF Generation

The PDF service uses ReportLab to assemble the title, summary, panel images, narration, dialogue, and page numbers into a downloadable comic document.

Frontend Development

The frontend uses HTML, CSS, and JavaScript. It provides the creation form, generation controls, loading status, comic preview, panel cards, panel regeneration, PDF download, and new-comic functionality. Fetch API is used to communicate with FastAPI without unnecessary page reloads.

Configuration and Utilities

python-dotenv is used for environment configuration. File utilities support safe file handling. API keys remain in .env rather than source code, and the application validates inputs and uses safe filenames and paths.

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